

Progress Report

NG-RC / NVAR Forecasting on the MIT-BIH ECG

Ahmed Khader

4 August 2026

1. What Was Done

Working from the existing per-patient preprocessing (100,000 samples of the MLI lead per patient, split **40k/10k/50k** into train/validation/test, MinMax-normalised with the scaler fit on train only, and windowed as **10 lags** $\rightarrow$ **next step**), I added two modelling stages.

Single-step NG-RC / NVAR

Recursive multi-step forecasting. To forecast further ahead, each single-step prediction is fed back in as the newest lag and the model is rolled forward to the horizon (the *recursive* strategy).

2. Results

All results are for patient 103. Single-step prediction is compared against a naive persistence baseline (predict the next value = the last observed value).

Table 1: Single-step forecasting (patient 103, test split).

Metric	NG-RC / NVAR	Naive baseline
RMSE	0.0054	0.0278
MAE	0.0040	—

Table 2: Recursive multi-step forecasting (patient 103, test split, over all samples).

Horizon	RMSE	MAE	R^2	Time (s)
10	0.0716	0.0301	0.619	0.10
20	0.1012	0.0499	0.239	0.21
50	0.1137	0.0589	0.039	0.52

3. Findings

- **Single-step is excellent.** At one step ahead the NVAR reaches $\text{RMSE} \approx 0.005$ on normalised data, about $5\times$ better than the persistence baseline.
- **Multi-step degrades fast.** Accuracy falls sharply with the horizon: R^2 drops from 0.62 (H=10) to 0.24 (H=20) to ≈ 0 (H=50). By 50 steps the model is barely better than predicting the flat average, because the recursion smooths away the sharp QRS peaks.

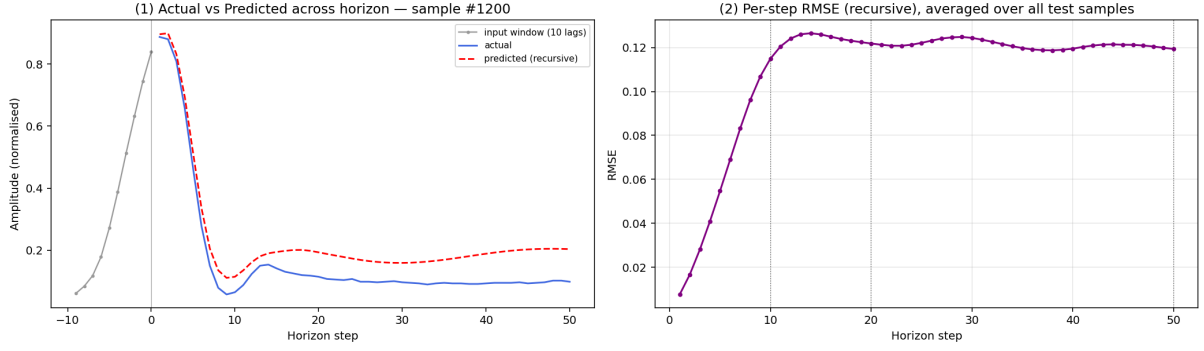


Figure 1: Recursive multi-step NG-RC/NVAR on patient 103. *Left:* actual versus predicted across the horizon for one sample, the forecast tracks the signal for the first few steps, then drifts. *Right:* per-step RMSE averaged over all test samples, showing how the error grows with horizon and then saturates.

4. Future Work / What to Add Next

- **Direct (or MIMO) multi-step** instead of recursive: train a separate read-out per horizon, or output the whole horizon at once
- **Longer or strided delay embedding** spanning roughly a full heartbeat, to give the model more temporal context (a deliberate departure from the paper’s fixed 10 lags).
- **Full evaluation:** run all 21 patients and report mean $\pm$ standard deviation, and add reference baselines (a linear AR model, and the deep models from the literature) so the results have a yardstick.